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Vehicle

Abstract

A vehicle includes: an internal combustion engine; a rotating electric machine; a power storage device; first and second electric heating type catalyst for purifying exhaust gas; a bidirectional power converter which converts first AC power into DC power, and convert DC power into a second AC power; a plurality of relays, which are electrically connected to a first power path and a second power path and which at least selects one of and switch between parallel connection and series connection of the first electric heating type catalyst and the second electric heating type catalyst; and a controller configured to control ON and OFF of each of the relays.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2023-176808 filed in Japan on Oct. 12, 2023.

BACKGROUND

(2) The present disclosure relates to a vehicle.

(3) Japanese Laid-open Patent Publication No. 2017-193245 discloses the electric heating type catalyst for purifying the exhaust gas discharged from the internal combustion engine and a vehicle that receives power from the power storage device.

SUMMARY

(4) There is a need for providing a vehicle capable of supplying power from the bidirectional power converter to the electric heating type catalyst of the two systems.

(5) According to an embodiment, a vehicle includes: an internal combustion engine; a rotating

electric machine, which is a driving source for running the vehicle; a power storage device for storing power to be supplied to the rotating electric machine; a first electric heating type catalyst and a second electric heating type catalyst, each of which has a catalyst for purifying exhaust gas discharged from the internal combustion engine and is supplied power from the power storage device to electrically heat the catalyst; and a bidirectional power converter having a DC port for electrically connecting to the power storage device, a first AC port to which a first AC power for charging the power storage device is input, and a second AC port for outputting a second AC power. Further, the bidirectional power converter that converts the first AC power into DC power to output the DC power to the DC port, and converts DC power supplied from the power storage device into the second AC power to output the second AC power to the second AC port, and the vehicle further includes: a plurality of relays, which are electrically connected, via a plurality of power lines, to a first power path, which connects the bidirectional power converter and the first AC port, and a second power path connecting the bidirectional power converter and the second AC port, that at least selects one of and switches between parallel connection and series connection of the first electric heating type catalyst and the second electric heating type catalyst; and a controller that controls ON and OFF of each of the relays.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter according to the embodiment;
(2) FIG. **2** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter in a parallel operation mode;
(3) FIG. **3** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter in a series operation mode.
(4) FIG. **4** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter in the first one-side operation mode.
(5) FIG. **5** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter in the second one-side operation mode;
(6) FIG. **6**A is a flow chart illustrating a half part of an exemplary control implemented by an ECU in the vehicles according to the embodiment; and
(7) FIG. **6**B is a flow chart illustrating another half part of the exemplary control implemented by the ECU in the vehicles according to the embodiment
DETAILED DESCRIPTION
(8) In a vehicle in which external charging to a power storage device and external power supply from a power storage device can be performed through a bidirectional power conversion device, there is room for investigation in technology for enabling power supply from the bidirectional power conversion device to two systems of electrically heated catalysts.
(9) Hereinafter, an embodiment of a vehicle according to the present disclosure. Note that the present disclosure is not limited by the present embodiment.
(10) The vehicle **100** according to the embodiment includes a bidirectional power converter **1**, a first Electrically Heated Catalyst (EHC) **21**, a second EHC **22**, and an Electronic Control unit (ECU) **30**, an inlet, an outlet, and a battery. The bidirectional power converter **1** includes a DC port for electrically connecting the bidirectional power converter **1** with a battery, a first AC port, a second AC port for outputting AC power, and a communication port **16** to which a communication line can be connected. The ECU **30** is, for example, a controller configured to control the bidirectional power converter **1**.
(11) When the vehicle performs external charging, AC power for charging the battery is inputted

into 1AC port. When the vehicle is powered by an electrical outlet, 2AC outlet provides AC power. When the car is powered externally, 1AC outputting AC power.

(12) The DC port is a connector for connecting a wire connected to the battery to the bi-directional power converter **1**. The first AC port is a connector for connecting a wire connected to an inlet to which the charge plug can be connected to the bidirectional power converter **1**. The second AC port is a connector for connecting a wire connected to an outlet for outputting AC power to the bidirectional power converter **1**. The bidirectional power converter **1** is detachably configured at each connector. Thus, replacement of the bidirectional power converter **1** is facilitated.

(13) The bidirectional power converter **1** is configured to convert the first AC power inputted to the first 1AC port during external charge to DC power and outputs it to DC port. Further, the bidirectional power converter **1**, at the time of outlet feeding, and is configured to convert the DC power supplied from the battery to DC port to the second AC power and outputs it to 2AC port. Further, the bidirectional power converter **1**, at the time of external power supply, and is configured to convert the DC power supplied from the battery to DC port to a third AC power and outputs it to the first AC port.

(14) FIG. **1** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter **1** according to the embodiment. As illustrated in FIG. **1**, the bidirectional power converter **1** according to the embodiment includes a capacitor **10**, an inverter **11**, an insulating circuit **12**, an inverter **13**, two reactor **14***a*, **14***b*, and a relay **15***a*, **15***b*, **15***c*, **15***d* and the like. The inverter **11** is a full bridge circuit including four switching elements of the first switching element **111** and the second switching element **112** and the third switching element **113** and the fourth switching element **114**. The insulating circuit **12** is an insulating transformer including a first coil **12***a* and the second coil **12***b*. The insulating circuit **12** performs transformation at a ratio corresponding to the turn ratio of the first coil **12***a* and the second coil **12***b*. The inverter **13** includes a first inverter **13***a*, a second inverter **13***b*, and a smoothing capacitor **13***c*. The first inverter **13***a* is a full bridge circuit including four switching elements of the first switching element **131***a* and the second switching element **132***a* and the third switching element **133***a* and the fourth switching element **134***a*. The second inverter **13***b* is a full bridge circuit including four switching elements of the first switching element **131***b* and the second switching element **132***b* and the third switching element **133***b* and the fourth switching element **134***b*. The switching elements included in the inverter **11** and the inverter **13**, ON and OFF is controlled by ECU **30**.

(15) In each of the power feeding and charging, the ECU **30** controls the bi-directional power conversion device **1** so that the bi-directional power conversion device **1** performs power conversion to be described below. Although not illustrated in FIG. **1**, in order to acquire the information used by the ECU **30** in the control, various sensors (e.g., a current sensor and a voltage sensor, etc.) are provided in an appropriate position of the circuit illustrated in FIG. **1**.

(16) At the time of power feeding, the inverter **11** is outputted to the insulating circuit **12** converts the DC power inputted from the first end (DC port-side) to a high-frequency AC power. The isolation circuitry **12** transfers the output of the inverter **11** (AC power) to the first inverter **13***a*. The first inverter **13***a* rectifies the AC power received from the insulating circuit **12** and outputs it to the second inverter **13***b*. At this time, the smoothing capacitor **13***c* is charged by the output power from the first inverter **13***a*, the output voltage of the first inverter **13***a* and substantially the same voltage. The second inverter **13***b* converts the DC power received from the first inverter **13***a* into a predetermined AC power to be outputted to the second end (reactor **14***a*, **14***b*).

(17) In charging, the second inverter **13***b* is outputted to the first inverter **13***a* rectifies the AC power inputted from the second end (reactor **14***a*, **14***b*). The first inverter **13***a* converts the DC power received from the second inverter **13***b* to AC power of high frequency. The isolation circuitry **12** transmits the output of the first inverter **13***a* (AC power) to the inverter **11**. The inverter **11** rectifies the AC power received from the insulating circuit **12** and outputs the first end (DC portside).

(18) The inlet is electrically connected to the first AC port of the bi-directional power converter **1** through a wire. A connector is provided at the end of the wire leading to the inlet, and the connector of the inlet is connected to the first AC port. The inlet is installed so that the user can use it from outside the vehicle. The inlet may be provided, for example, on the rear side of the body. The inlet is covered by a charging lid when not in use. The inlet is exposed when the user opens the charging lid. Then, by the user connecting the plug of the Electric Vehicle Supply Equipment (EVSE) to the inlet, the power can be supplied to the vehicles using EVSE. The plug of EVSE is, for example, a connector of a charging cable, and when the plug is connected to the inlet, EVSE and the vehicle are electrically connected via the charging cable. When powered externally, the user can power EVSE from the inlet of the vehicle.
(19) The outlet is electrically connected to the second AC port of the bi-directional power converter **1** through a wire. A connector is provided at the end of the wire leading to the outlet, and the connector of the outlet is connected to the second AC port. The outlet is a vehicle interior outlet installed in the vehicle cabin. The outlet outputs the second AC power.
(20) The second AC power is, for example, 100V AC power.
(21) The battery is electrically connected to DC of the bi-directional power converter **1** through a wire. The battery is a power storage device that stores electric power supplied to the rotating electric machine to be described later. The battery is, for example, a secondary battery such as a lithium ion battery or a nickel metal hydride battery. In the present embodiment, as the secondary battery, and employs a battery pack including a plurality of lithium ion batteries. The assembled battery is configured by a plurality of battery cells are electrically connected to each other.
(22) The ECU **30** is connected to a communication port **16** of the bidirectional power converter **1** via a signaling line. The ECU **30** is composed of a processor, a Random Access Memory (RAM), and a storage device. For example, a Central Processing Unit (CPU) can be employed as a processor. RAM functions as a working memory for temporarily storing data are to be processed by the processor. The storage device is configured to store stored information. As a storage device, for example, a Read Only Memory (ROM) and a rewritable non-volatile memory can be employed. In addition to the program, information used in the program (e.g., maps, mathematical expressions, and various parameters) is stored in the memory. In an embodiment, various controls in the ECU **30** are executed by executing a program stored in the storage device by the processor.
(23) The vehicles further include a Power Control Unit (PCU), a rotating electric machine, a power transmission gear, drive wheels, a System Main Relay (SMR), and a starter. The vehicle can run by supplying the electric power stored in the battery to the rotating electric machine which is a driving source for the vehicle running.
(24) The SMR is provided in the power path connecting the battery and the PCU. The connecting or disconnecting state of SMR is controlled by the ECU **30**. When the SMR is connected, power can be exchanged between the battery and PCU. On the other hand, when the SMR is shut off, it becomes impossible to transfer electric power between the battery and the PCU. The SMR is connected, for example, when the vehicle is running.
(25) The rotating electric machine is, for example, a three-phase AC motor generator. The rotating electric machine is driven by the PCU and is configured to rotate the driving wheels of the vehicles. The PCU includes, for example, inverters and converters. The PCU inverters and converters are controlled by the ECU **30**. The output torque of the rotating electricity is transmitted to the drive wheels through a power transmission gear that serves as a reduction gear or differential. The rotating electric machine is also configured to perform regenerative power generation and supply the generated power to the battery.
(26) The activation switch is a switch for starting the vehicle system. The activation switch is operated when the system is stopped, thereby activating the vehicular system including the ECU **30**. On the other hand, the vehicle system is stopped by the activation switch is operated when the system is activated. The start-up switch is commonly referred to as a power switch or ignition

switch.
(27) In this embodiment, the power path connecting the bi-directional power converter **1** and the first AC port is referred to as a first power path. A power line PL**1***a* constitutes a part of the first power path of the first polarity, and power line PL**1***b* constitutes a part of the first power path of the second polarity. Further, in the present embodiment, the power path connecting the bidirectional power converter **1** and the second AC port is referred to as a second power path. A power line PL**2***a* constitutes a part of the second power path of the first polarity, and the power line PL**2***b* constitutes a part of the second power path of the second polarity. The first power path and the second power path are selectively switchable by a switching device having a plurality of relays.
(28) The first end of the bidirectional power converter **1**, the power line PL**11***a* of the first polarity and the power line PL**11***b* of the second polarity are connected. The second end of the bidirectional power converter **1**, the power line PL**12***a* of the first polarity and the power line PL**12***b* of the second polarity are connected. In charging, power is input to the second end side, power is output from the first end side. At the time of power feeding, the power is input to the first end side, power is output from the second end side. The first polarity and the second polarity are opposite polarities.
(29) Each of the power lines PL**11***a*, PL**11***b* is connected to a DC porting. The power line PL**12***a* is branched into a power line PL**1***a* and a power line PL**2***a* at the first branch point P**1**. To the power line PL**1***a*, the first relay **15***a* is connected. To the power line PL**2***a*, the third relay **15***c* is connected. The power line PL**12***b* is branched into a power line PL**1***b* and a power line PL**2***b* at the second branch point P**2**. To the power line PL**1***b*, the second relay **15***b* is connected. To the power line PL**2***b*, the fourth relay **15***d* is connected.
(30) The first EHC **21** is connected to the first relay **15***a* by a power line PL**3** and is connected to the fourth relay **15***d* by a power line PL**4** and a power line PL**5**. The power line PL**4** is branched into a power line PL**5** and a power line PL**7** at the third branch point P**3**. The second EHC **22** is connected to the second relay **15***b* by a power line PL**6** and is connected to the fourth relay **15***d* by a power line PL**7** and a power line PL**5**. The third relay **15***c* is connected to the power line PL**6** at the fourth branch point P**4** by a power line PL**8**.
(31) The first EHC **21** and the second EHC **22** are electrically heated catalysts provided in the exhaust passage of the engine, which is an internal combustion engine provided in the vehicle. The first EHC **21** and the second EHC **22** has a catalyst converter for purifying the exhaust gases emitted from the engine. The first EHC **21** and the second EHC **22**, for example, are configured to electrically heat the catalyst converter by being supplied with power from a battery through the bidirectional power converter **1**.
(32) In the vehicle according to the embodiment, the bi-directional power converter **1** has a function as a power supply device of the first EHC **21** and the second EHC **22**. When the bi-directional power converter **1** functions as a power supply device of the first EHC **21** and the second EHC **22**, for example, a smoothing capacitor **13***c* of the inverter **13** as a power source, a battery (the first inverter **13***a* of the inverter **13**) It supplies power to at least one of the first EHC **21** and the second EHC **22** from the smoothing capacitor **13***c* charged by the output power. In the vehicle according to the embodiment, for example, the power output from the first inverter **13***a* of the inverter **13**, the output power of the power source when the bi-directional power converter **1** may function as a power supply of the first EHC **21** and the second EHC **22**. Further, in the vehicle according to some embodiments, as a configuration in which the bidirectional power converter **1** functions as a power supply device of the first EHC **21** and the second EHC **22**, the second inverter **13***b* of the inverter **13** is not required.
(33) In the vehicle according to the embodiment, to a first power path connecting the bidirectional power converter **1** and the first AC port, and a second power path connecting the bidirectional power converter **1** and second AC port via a plurality of power lines, the first EHC **21** and the second EHC **22** are electrically connected. Furthermore, in the vehicle according to the embodiment, the parallel connection and the series connection of the first EHC **21** and the second

EHC **22** as a plurality of relays which can be at least selectively switched, the first relay **15***a* and the second relay **15***b* and four relays of the third relay **15***c* and the fourth relay **15***d* are electrically connected to such a plurality of power lines.

(34) In the vehicle according to the embodiment, the control of ON and OFF of each of the first relay **15***a* and the second relay **15***b* and the third relay **15***c* and the fourth relay **15***d* is performed by the ECU **30**. When the relays are turned ON, the relays are closed to enable energization. When the relays are turned OFF, the relays are opened so that the power cannot be turned on. Thus, the ECU **30** is capable of selective switching of the four operation modes of the parallel operation mode and the series operation mode and the first one-side operation mode and the second one-side operation mode. In the parallel operation mode, the first EHC **21** and the second EHC **22** are connected in parallel to operate the first relay **15***a*, the second relay **15***b*, the third relay **15***c*, and the fourth relay **15***d*, so that power can be supplied to the first EHC **21** and the second EHC **22**. In series operation, the second EHC **21** and the second EHC **22** are connected in series to operate the first relay **15***a*, the second relay **15***b*, the third relay **15***c*, and the fourth relay **15***d* so that the power can be supplied to the first EHC **21** and the second EHC **22**. In the first one-side operation mode, so as to be capable of supplying power only to the first EHC **21**, operates the first relay **15***a* and the second relay **15***b* and the third relay **15***c* and the fourth relay **15***d*. In the second one-side operation mode, so as to be capable of supplying power only to the second EHC **22**, operates the first relay **15***a* and the second relay **15***b* and the third relay **15***c* and the fourth relay **15***d*.

(35) FIG. **2** is a diagram illustrating an example of a circuit configuration of a bidirectional power converter in a parallel operation mode.

(36) As illustrated in FIG. **2**, in the parallel operation mode, the first relay **15***a*, the third relay **15***c*, and the fourth relay **15***d* are turned ON, the second relay **15***b* is turned OFF, and the first EHC **21** and the second EHC **22** are connected in parallel to supply power to the first EHC **21** and the second EHC **22**. Further, in the parallel operation mode, the second inverter **13***b* of the inverter **13**, the second switching element **132***b* and the third switching element **133***b* to ON, the first switching element **131***b* and the fourth switching element **134***b* to OFF.

(37) Thick arrows in FIG. **2**, as viewed from the smoothing capacitor **13***c* of the inverter **13** as a power supply, illustrates the flow of current returning from the high voltage side of the smoothing capacitor **13***c* to the low-voltage side of the smoothing capacitor **13***c* through the first EHC **21** and the second EHC **22**.

(38) As illustrated in FIG. **2**, in the parallel operation mode, the current flowing from the smoothing capacitor **13***c* to the power line PL**12***a* through the third switching element **133***b* and the reactor **14***a*, the power line PL**1***a* at the first branch point P**1** and the power line PL**2***a* branch.

(39) The current branched to the power line PL**1***a* at the first branch point P**1** is supplied to the first EHC **21** via the first relay **15***a* and the power line PL**3**. Then, the current flowing from the first EHC **21** to the power line PL**2***b* through the power line PL**4** and the power line PL**5** and the fourth relay **15***d* flows to the power line PL**12***b* at the second branch point P**2**. The current flowing to the power line PL**12***b* at the second branch point P**2** returns to the smoothing capacitor **13***c* via the reactor **14***b* and the second switching device **132***b*.

(40) On the other hand, the current branched to the power line PL**2***a* side at the first branch point P**1** flows to the power line PL**6** side at the fourth branch point P**4** via the third relay **15***c* and the power line PL**8**. The current flowing to the power line PL**6** at the fourth branch point P**4** is supplied to the second EHC **22**. Then, it flows from the second EHC **22** through the power line PL**7** to the power line PL**5** at the third branch point P**3**. The current flowing to the power line PL**2***b* through the fourth relay **15***d* flows to the power line PL**5** at the third branch point P**3** flows to the power line PL**12***b* at the second branch point P**2**. The current flowing to the power line PL**12***b* at the second branch point P**2** returns to the smoothing capacitor **13***c* via the reactor **14***b* and the second switching device **132***b*.

(41) FIG. **3** is a diagram showing an example of a circuit configuration of a bidirectional power

converter **1** in a series operation mode.
(42) As illustrated in FIG. **3**, in the series operation mode, the first relay **15***a* and the second relay **15***b* are turned ON, the third relay **15***c* and the fourth relay **15***d* are turned OFF, and the first EHC **21** and the second EHC **22** are connected in series to provide power to the first EHC **21** and the second EHC **22**. Also in the series operation mode, the second inverter **13***b* of the inverter **13**, the second switching element **132***b* and the third switching element **133***b* to ON, the first switching element **131***b* and the fourth switching element **134***b* to OFF.
(43) The thick arrows in FIG. **3**, as viewed from the smoothing capacitor **13***c* of the inverter **13** as a power supply, indicates the flow of current returning to the low-voltage side of the smoothing capacitor **13***c* through the first EHC **21** and the second EHC **22** from the high-voltage side of the smoothing capacitor **13***c*.
(44) As illustrated in FIG. **3**, in the series operation mode, the current flowing from the smoothing capacitor **13***c* to the power line PL**12***a* through the third switching element **133***b* and the reactor **14***a*, the power line PL**1***a* at the first branch point P**1** It flows to.
(45) The current flowing to the power line PL**1***a* at the first branch point P**1** is supplied to the first EHC **21** via the first relay **15***a* and the power line PL**3**. Then, the current flowing from the first EHC **21** through the power line PL**4** to the power line PL**7** in the third branch point P**3** is supplied to the second EHC **22**. Then, the current flowing from the second EHC **22** to the power line PL**6** flows to the power line PL**9** at the fourth branch point P**4**. The current flowing to the power line PL**9** side at the fourth branch point P**4** flows to the power line PL**12***b* side at the second branch point P**2** via the second relay **15***b* and the power line PL**1***b*. The current flowing to the power line PL**12***b* at the second branch point P**2** returns to the smoothing capacitor **13***c* via the reactor **14***b* and the second switching device **132***b*.
(46) FIG. **4** is a diagram illustrating an example of the circuit configuration of the bidirectional power converter **1** in the first one-side operation mode.
(47) As illustrated in FIG. **4**, in the first one-side operation mode, the first relay **15***a* and the fourth relay **15***d* are turned ON, the second relay **15***b* and the third relay **15***c* are turned OFF, and only the first EHC **21** is operated by supplying power to the first EHC **21**. Also in the first one-side operation mode, the second inverter **13***b* of the inverter **13**, the second switching element **132***b* and the third switching element **133***b* to ON, the first switching element **131***b* and the fourth switching element **134***b* to OFF.
(48) The thick arrows in FIG. **4**, as viewed from the smoothing capacitor **13***c* of the inverter **13** as a power supply, indicates the flow of current returning to the low-voltage side of the smoothing capacitor **13***c* through the first EHC **21** from the high-voltage side of the smoothing capacitor **13***c*.
(49) As illustrated in FIG. **4**, in the first one-side operation mode, the current flowing from the smoothing capacitor **13***c* to the power line PL**12***a* through the third switching element **133***b* and the reactor **14***a*, the power line PL**1***a* at the first branch point P**1** it flows to.
(50) The current flowing to the power line PL**1***a* at the first branch point P**1** is supplied to the first EHC **21** via the first relay **15***a* and the power line PL**3**. Then, the current flowing from the first EHC **21** to the power line PL**2***b* through the power line PL**4** and the power line PL**5** and the fourth relay **15***d* flows to the power line PL**12***b* at the second branch point P**2**. The current flowing to the power line PL**12***b* at the second branch point P**2** returns to the smoothing capacitor **13***c* via the reactor **14***b* and the second switching device **132***b*.
(51) FIG. **5** is a diagram showing an example of the circuit configuration of the bidirectional power converter **1** in the second one-side operation mode.
(52) As illustrated in FIG. **5**, in the second one-side operation mode, the third relay **15***c* and the fourth relay **15***d* are turned ON, the first relay **15***a* and the second relay **15***b* are turned OFF, and only the second EHC **22** is operated by supplying power to the second EHC **22**. Also in the second one-side operation mode, the second inverter **13***b* of the inverter **13**, the second switching element **132***b* and the third switching element **133***b* to ON, the first switching element **131***b* and the fourth

switching element **134***b* to OFF.
(53) The thick arrows in FIG. **5**, as viewed from the smoothing capacitor **13***c* of the inverter **13** as a power supply, indicates the flow of current returning to the low-voltage side of the smoothing capacitor **13***c* through the second EHC **22** from the high-voltage side of the smoothing capacitor **13***c*.
(54) As illustrated in FIG. **5**, in the second one-side operation mode, the current flowing from the smoothing capacitor **13***c* to the power line PL**12***a* through the third switching element **133***b* and the reactor **14***a* flows to the power line PL**2***a* at the first branch point P**1**.
(55) The current flowing to the power line PL**2***a* side at the first branch point P**1** flows to the power line PL**6** side at the fourth branch point P**4** via the third relay **15***c* and the power line PL**8**. The current flowing to the power line PL**6** at the fourth branch point P**4** is supplied to the second EHC **22**. Then, the current flowing from the second EHC **22** to the power line PL**7** flows to the power line PL**5** at the third branch point P**3**. The current flowing to the power line PL**5** side at the third branch point P**3** flows to the power line PL**12***b* side at the second branch point P**2** via the fourth relay **15***d* and the power line PL**2***b*. The current flowing to the power line PL**12***b* at the second branch point P**2** returns to the smoothing capacitor **13***c* via the reactor **14***b* and the second switching device **132***b*.
(56) Here, in the parallel operation mode, when the respective resistance values of the first EHC **21** and the second EHC **22** are scattered, for example, power supplied to the lower EHC of the resistance value will be concentrated, there is a possibility that differences occur in the warm air in the first EHC **21** and the second EHC **22**. Further, in the series operation mode, since the output upper limit voltage becomes low when the input (battery side) voltage is low, cannot be sufficiently applied voltage to the first EHC **21** and the second EHC **22** connected in series, the warm-up performance there is a risk of deterioration.
(57) FIGS. **6**A and **6**B provide a flow chart illustrating an exemplary control implemented by the ECU **30** in the vehicles according to the embodiment. Incidentally, in FIGS. **6**A and **6**B, Vout represents the power supply output-voltage. Further, in FIGS. **6**A and **6**B, Iout represents power output current, in other words, represents the supplied current of either or both of the first EHC **21** and the second EHC **22**. Further, in FIGS. **6**A and **6**B, Imax represents the allowable current upper limit of either or both of the first EHC **21** and the second EHC **22** (EHC hardware requirements). Further, in FIGS. **6**A and **6**B, Vmax represents the voltage upper limit that the power supply can output (change in the input voltage, etc.).
(58) First, The ECU **30** is activated in step S**1** by the activation-switch being operated when the system is stopped. The ECU **30** is then set to parallel operation in step S**2**. Next, The ECU **30**, in step S**3**, from the output voltage and the output current to the first EHC **21** and the second EHC **22**, calculates a resistance-value R**1**, R**2** of the first EHC **21** and the second EHC **22**. Next, the ECU **30** determines whether or not the relationship of Vmax>Imax(R**1**+R**2**) is satisfied in step S**4**. That is, when it is possible to supply up to the current upper limit in the series operation mode, priority is given to the series operation mode in order to avoid power concentration. The ECU **30**, in step S**4**, when it is determined that does not satisfy the relationship of Vmax>Imax (R**1**+R**2** (No at step S**4**), the process proceeds to step S**5**. The ECU **30** continues to operate in parallel in step S**5**.
(59) The ECU **30** then determines, in step S**6**, whether any of EHC satisfy Iout>Imax relationship. The ECU **30**, in step S**6**, when it is determined that satisfies Iout>Imax relationship (Yes at step S**6**), the process proceeds to step S**7**. The ECU **30** then lowers Vout in step S**7** and moves to step S**6**. Further, when the ECU **30**, in step S**6**, determines that any EHC does not satisfy Iout>Imax relationship (No at step S**6**), the process proceeds to step S**15**.
(60) Further, when the ECU **30**, in step S**4**, determines that the relationship of Vmax>Imax(R**1**+R**2**) satisfies (Yes at step S**4**), the process proceeds to step S**8**. The ECU **30** switches to series operation in step S**8**. At this time, the ECU **30** is outputted by Vout=Imax(R**1**+R**2**). The ECU **30** then determines whether or not Iout>Imax relationship is satisfied in step S**9**. The ECU **30**, in step S**9**,

determines that Iout>Imax relationship is satisfied (Yes at step S**9**), the process proceeds to step S**10**. The ECU **30** lowers Vout at the step S**10** and moves to the step S**9**. Further, the ECU **30**, in step S**9**, determines that Iout>Imax relationship is not satisfied (No at step S**9**), the process proceeds to step S**11**.

(61) In addition, the ECU **30** determines whether or not to perform Vout limit in other than EHC requirement in the step S**11**. In other words, when other requirements other than EHC requirements must be used to lower the voltage. If the full output cannot be achieved, the process returns to the parallel operation mode. When the ECU **30**, in step S**11**, determines to execute EHC limit in other than Vout requirement (Yes at step S**11**), the process proceeds to step S**12**. In step S**12**, the ECU **30** determines whether the relationship of Vmax>Imax(R**1**+R**2**) is satisfied. The ECU **30**, in step S**12**, determines that the relationship of Vmax>Imax(R**1**+R**2**) is not satisfied (No at step S**12**), the process proceeds to step S**13**. The ECU **30** switches to the parallel operation mode in step S**13**, and shifts to the step S**3**.

(62) Further, the ECU **30**, in step S**12**, determines that the relationship of Vmax>Imax(R**1**+R**2**) is satisfied (Yes at step S**12**), the process proceeds to step S**14**. In step S**14**, the ECU **30** outputs Vout=Iout(R**1**+R**2**), and the process proceeds to step S**15**.

(63) In the step ECU **30**, in step S**11**, when it is determined not to execute EHC limit other than Vout limit (No at step S**11**), the process proceeds to step S**15**.

(64) The ECU **30** determines whether or not the supplied electric energy to the first EHC **21** or the second EHC **22** is equal to or more than a threshold value in step S**15**. That is, when the desired power supply to one of the EHCs is completed, the power supply to only the other EHC. When the ECU **30**, at step S**15**, determines that the supplied power to the first EHC **21** or the second EHC **22** is not to be equal to or greater than the threshold value (No at step S**15**), the process proceeds to step S**16**. The ECU **30** determines whether or not it is currently in parallel operation-mode in step S**16**. When the ECU **30** determines, in step S**16**, at present, that the parallel operation mode (Yes at step S**16**), the process proceeds to step S**3**. Further, the ECU **30**, in step S**16**, at present, determines that it is not the parallel operation mode (No at step S**16**), the process proceeds to step S**9**.

(65) Further, the ECU **30** determines, in step S**15**, that the supplied power to the first EHC **21** or the second EHC **22** is equal to or greater than the threshold value (Yes at step S**15**), the process proceeds to step S**17**. When the ECU **30** determines whether the EHC in which the supplied electric energy reaches the threshold value is the first EHC **21** in step S**17**.

(66) When the ECU **30** determines, in step S**17**, that the EHC in which the supplied power amount has reached the threshold value is the first EHC **21** (Yes at step S**17**), the process proceeds to step S**18**. The ECU **30** switches to the second one-sided mode in step S**18**. Next, the ECU **30** determines whether or not the energizing current is equal to or less than the allowable value in step S**19**. In step S**19**, the ECU **30** proceeds to step S**20** and determines that the energization current is not less than the allowable value (No at step S**19**). The ECU **30**, in step S**20**, performs a voltage-adjustment in the switching control in the second inverter **13***b* or the like of the inverter **13**, and proceeds to step S**19**.

(67) Further, when the ECU **30**, in step S**19**, determines that the energization current is equal to or less than the allowable value (Yes at step S**19**), the process proceeds to step S**21**. The ECU **30** determines whether or not the power supplied to the second EHC **22** is equal to or more than a threshold value in step S**21**. When the ECU **30**, in step S**21**, determines that the supplied power to the second EHC **22** is not equal to or greater than the threshold value (No at step S**21**), the process proceeds to step S**19**. Further, when the ECU **30** determines, in step S**21**, that the supplied power to the second EHC **22** is equal to or greater than the threshold value (Yes at step S**21**), the process proceeds to step S**22**. The ECU **30** performs a power outage (stop output) in step S**22**. The ECU **30** then terminates the series of controls.

(68) In addition, when the ECU **30**, in step S**17**, determines that the EHC in which the supplied electric energy has reached the threshold value is not the first EHC **21** but the second EHC **22** (No

at step S**17**), the process proceeds to step S**23**. The ECU **30** switches the relay to the first one-side-mode in step S**23**. Next, the ECU **30** determines whether or not the energizing current is equal to or less than an allowable value in step S**24**. In step S**24**, the ECU **30** proceeds to step S**25** when it is determined that the energization current is not less than the allowable value (No at step S**24**). The ECU **30**, in step S**25**, performs a voltage-adjustment in the switching control in the second inverter **13***b* or the like of the inverter **13**, the process proceeds to step S**24**.
(69) Further, when the ECU **30** determines, in step S**24**, that the energization current is equal to or less than the allowable value (Yes at step S**24**), the process proceeds to step S**26**. The ECU **30** determines whether or not the power supplied to the first EHC **21** is equal to or more than a threshold value in step S**26**. When the ECU **30** determines, in step S**26**, that the supplied power to the first EHC **21** is not equal to or greater than the thresholds (No at step S**26**), the process proceeds to step S**24**. Further, when the ECU **30** determines, in step S**26**, that the supplied power to the first EHC **21** is be equal to or greater than the threshold (Yes at step S**26**), the process proceeds to step S**22**. The ECU **30** performs a power outage in step S**22**. The ECU **30** then terminates the series of controls.
(70) In the vehicles according to the embodiment, the power can be supplied to the EHCs of the two systems (i.e., the first EHC **21** and the second EHC **22**) with one power source. Further, in the vehicle according to the embodiment, the control of ON and OFF of the relays **15***a*, **15***b*, **15***c*, and **15***d* by the ECU **30** enables the selective switching of the four operation modes of the parallel operation mode, the series operation mode, the first one-side operation mode, and the second one-side operation mode, thereby enabling efficient power to be supplied to the first EHC **21** and the second EHC **22**. In addition, the control of ON and OFF of the relays **15***a*, **15***b*, **15***c*, and **15***d* can simultaneously provide power to the first EHC **21** and the second EHC **22** and power to 2AC outlet and the electrically connected in-vehicle outlet in any operation mode.
(71) Vehicle according to the present disclosure has an effect that it is possible to supply power from the bidirectional power conversion device to the electric heating type catalyst of the two systems of the first electric heating type catalyst and the second electric heating type catalyst.
(72) According to an embodiment, it is possible to supply power from the bidirectional power conversion apparatus to the first electric heating type catalyst and the second electric heating type catalyst two systems of the electric heating type catalyst.
(73) According to an embodiment, it is possible to efficiently power supply to the first electric heating type catalyst and the second electric heating type catalyst.
(74) Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A vehicle comprising: an internal combustion engine; a rotating electric machine, which is a driving source for running the vehicle; a power storage device for storing power to be supplied to the rotating electric machine; a first electric heating type catalyst and a second electric heating type catalyst, each of which has a catalyst for purifying exhaust gas discharged from the internal combustion engine and is configured to be supplied power from the power storage device to electrically heat the catalyst; and a bidirectional power converter having a DC port for electrically connecting to the power storage device, a first AC port to which a first AC power for charging the power storage device is input, and a second AC port for outputting a second AC power, wherein the bidirectional power converter is configured to convert the first AC power into DC power to output the DC power to the DC port, and convert DC power supplied from the power storage device into the second AC power to output the second AC power to the second AC port, and the vehicle further

comprising: a plurality of relays, which are electrically connected, via a plurality of power lines, to a first power path, which connects the bidirectional power converter and the first AC port, and a second power path connecting the bidirectional power converter and the second AC port, configured to at least select one of and switch between parallel connection and series connection of the first electric heating type catalyst and the second electric heating type catalyst; and a controller configured to control ON and OFF of each of the relays.

2. The vehicle according to claim 1, wherein the controller is configured to control ON and OFF of each of the relays to selectably switch among a parallel operation mode, a series operation mode, a first one-side operation mode, and a second one-side operation mode, wherein in the parallel operation mode, the first electric heating type catalyst and the second electric heating type catalyst are connected in parallel and power is supplied to both the first electric heating type catalyst and the second electric heating type catalyst; in the series operation mode, the first electric heating type catalyst and the second electric heating type catalyst are connected in series and power is supplied to both the first electric heating type catalyst and the second electric heating type catalyst; in the first one-side operation mode, power is supplied only to the first electric heating type catalyst; and in the second one-side operation mode, power is supplied only to the second electric heating type catalyst.

3. The vehicle according to claim 1, wherein the rotating electric machine is a motor generator and the power storage device is a battery.

4. The vehicle according to claim 2, wherein the rotating electric machine is a motor generator and the power storage device is a battery.